# Dominick George

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## PROFILE

Software engineer and MSc Cyber Security candidate with three years of client-facing, Agile project delivery at Cardiff University's National Software Academy. Builds full-stack web, mobile and cloud applications in Java, Python and JavaScript, and is now specialising in penetration testing, application security and digital forensics on an NCSC-certified programme co-developed with PwC. Seeking graduate roles in software engineering, application security or security engineering in the UK from July 2027.

## EDUCATION

**MSc Cyber Security and Technology** — Cardiff University · Sep 2026 – July 2027 (expected)

- NCSC-certified, BCS-accredited programme developed with PwC's Ethical Hacking and Cyber Security team.
- Modules: Penetration Testing and Malware Analysis; Application and System Security; Computer and Network Forensics; Operating Systems Security; Applied Cryptography and Infrastructure Security; Cyber Security and Risk Management.

**BSc (Hons) Applied Software Engineering** — Cardiff University, National Software Academy · Sep 2023 – Jun 2026 · (2.1)

- BCS-accredited, industry-designed degree built around four-week client projects for private, public and third-sector organisations.
- Key modules: Commercial Applications with Java, DevOps, Cyber Security, Performance and Scalability, Mobile Development, Agile Project Management, Database Systems, Emerging Technologies.
- Final-year Large Team Project: delivered TaxAlly for an investor over 10 weeks as part of a 5-person team.

## TECHNICAL SKILLS

| Area | Skills |
|---|---|
| **Languages** | Java, Python, JavaScript, SQL, Kotlin |
| **Web and mobile** | Spring Boot, React, Node.js, REST APIs, HTML/CSS |
| **Data** | Relational (MySQL & PostgreSQL) and NoSQL (MongoDB) design, query tuning, database security |
| **DevOps and cloud** | Git, GitLab CI/CD, Docker, Azure, automated testing, performance and load testing |
| **Security** | Penetration testing, network enumeration and port scanning, malware analysis, OWASP Top 10 (SQLi, XSS, CSRF), OS hardening, applied cryptography, secure cloud configuration |
| **Security tools** | Nmap, Burp Suite, Wireshark, Metasploit, Kali Linux, Autopsy / Volatility |
| **Ways of working** | Agile / Scrum, requirements gathering with clients, code review, sprint planning, risk assessment |

## SELECTED PROJECTS

**TaxAlly** · Final-year team project · Next.js, Django REST, PostgreSQL · Project page

- Built authentication end to end for an AI-assisted UK Self-Assessment platform: JWTs in HttpOnly cookies, CSRF protection, token refresh and protected routes.
- Delivered sign-up with client- and server-side validation, consent-gated registration and a privacy-preferences model, plus an Insights dashboard annualising Income Tax and National Insurance.
- Wrote backend unit tests and API end-to-end tests with pytest, within a Docker and Jenkins pipeline that runs Bandit, Safety and npm audit scans.

**NHS Wales Recruitment App** · University client project · Spring Boot, Spring Security · Project page

- Built an applicant tracking system for NHS Wales in a five-person Agile team, meeting the client regularly and presenting the final product to them.
- Spring Security protects every route, with BCrypt password hashing, role-based access, CSRF protection and hashed, expiring password-reset tokens.

- Tested controllers, services and repositories with Spring Boot Test.

**Accessible Gaming Platform** · Team project · React, NestJS, PostgreSQL · Project page

- Owned security hardening for a platform helping disabled players find accessible mobile games: locked-down CORS, security headers, a Content Security Policy and validated, bounded queries.
- Built the admin dashboard for verifying games, managing accessibility tags and resolving user-reported issues, with unit and end-to-end tests.
- Set up CI/CD: GitLab CI tests with coverage, plus Terraform that provisions an OpenStack VM running Jenkins to test, build and deploy the Docker stack.

**Jarvis: Private AI Assistant** · Personal project · Python, llama.cpp, Whisper · Project page

- Built a voice-driven assistant for my notes that runs a 35-billion-parameter open model entirely on a 12 GB consumer GPU, calibrated to 5.9 GB of VRAM at about 23 tokens a second.
- Designed confirm-before-acting Gmail and Calendar actions with prompt-injection defences, one-time action IDs and least-privilege OAuth scopes (drafts, never sends).
- Verified with automated tests: 15 of 15 action and prompt-injection prompts handled correctly, and no requests leaving the machine.

**Secure Web Server** · Lab work · Python, Diffie–Hellman, AES · Project page

- Built a TCP client and server that agree an AES-128 key through a Diffie–Hellman exchange written from the maths up, validating the public parameters and using secure randomness.
- Documented the path to production: 2048-bit groups or X25519, AES-GCM with HKDF, and an authenticated key exchange to stop man-in-the-middle attacks.

More work: Portfolio · GitHub

## EXPERIENCE

**Head of Path Planning** — Cardiff Autonomous Racing, Cardiff University · Project page

- Lead the path planning team of eight engineers as a founding member, working through trade-offs with other team leads ahead of competing at Silverstone.
- Built the team's path-planning ROS 2 node in Python: it pairs SLAM odometry with YOLO cone detections, builds the track boundaries and publishes a minimum-curvature racing line to the control team at 5 Hz.
- Created a Tkinter/Matplotlib tester for scenario-based testing and traced failing scenarios to duplicate cones breaking the spline fit.

## CERTIFICATIONS AND ACTIVITIES

- Hack The Box Academy: completed courses in Web Requests, Introduction to Web Applications and Introduction to Linux.
- Business Transformation in the Age of AI (conference, Bristol): heard industry leaders discuss emerging technologies, networked with key figures from major companies and gained insight into how businesses are adopting AI.
- Independent learning: online courses, coding bootcamps and self-directed projects, including a personal portfolio website, a secure login system and a 2D Java RPG game.

References available on request.